GNSS Position Time Series Analysis

Quantifying GNSS velocity uncertainty under realistic (correlated) noise - Cascades region, 4 EarthScope stations.

Chelsea Momoh
Statistics, UC Davis, 2026

Table of contents

Introduction

The project has three parts:

- Signal Decomposition
- Pruned Exact Linear Time (PELT) Changepoint Detection
- Uncertainty Quantification (Covariance Realism)

The project culminates in the quantification of the unreliability of Ordinary Least Squares' stated uncertainty.

What Will This Presentation Lead to?

Motivating Question: Can we quantify how erroneous Ordinary Least Squares' formal uncertainty is when assumptions are violated?

- Compare to Generalized Least Squares' formal uncertainty
- Use Monte Carlo simulated "ground truth"

01

The Data

The Cascades

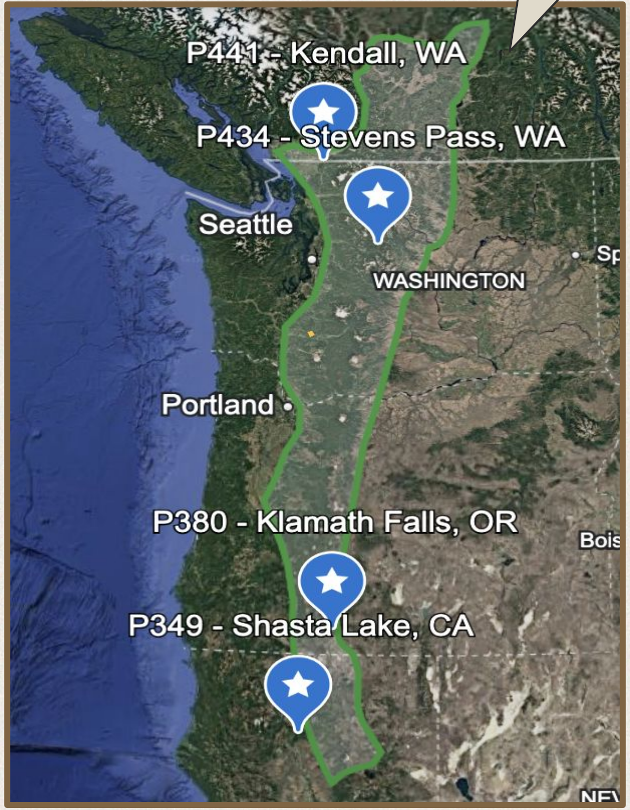
Four Ground Stations

Reference Frame:
- Nam14
  - The newest North American fixed reference frame
  - Removes plate motion from displacement data

Ground Stations:
- P441, P434, P380, P349

Data:
- Tracks displacement (mm) over time
- Sampling Rate: Once per day
- Data length: 15-20 years (6000-8000 rows)
  - ~2005–2026

Click Me!



Sourced [From Earthscope]

02

Signal Decomposition

Model Construction

The Model

position(t) = a + bt + c·sin(2πt) + d·cos(2πt) + e·sin(4πt) + f·cos(4πt) + residual [1]

a - initial offset
b - tectonic velocity (mm/year)
c, d - annual seasonal signal (snow loading, hydrology)
e, f - semi-annual seasonal signal
t - decimal years since first observation

Construction

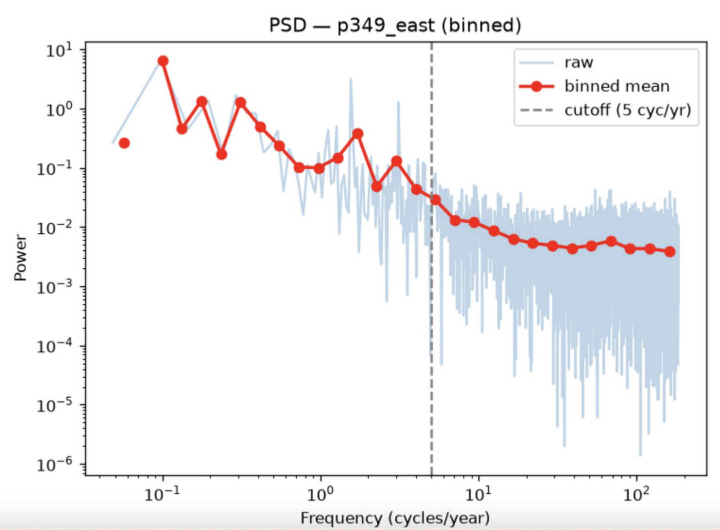
Built by hand in Numpy

1. Montillet and Bos, "7.2.1 Least Squares Estimation," essay, in *Geodetic Time Series Analysis in Earth Sciences* (Springer, n.d.), 215.

Spectral Analysis / Noise Characterization

Method

- 12 PSD graphs via scipy.signal periodogram
- Translates noise signal to the frequency domain
- Reveals the noise isn't white / The noise is correlated / The noise is not independent.
- Means OLS' assumption of independent noise is violated



Power Spectral Density (PSD) graph of station 349's east direction

03

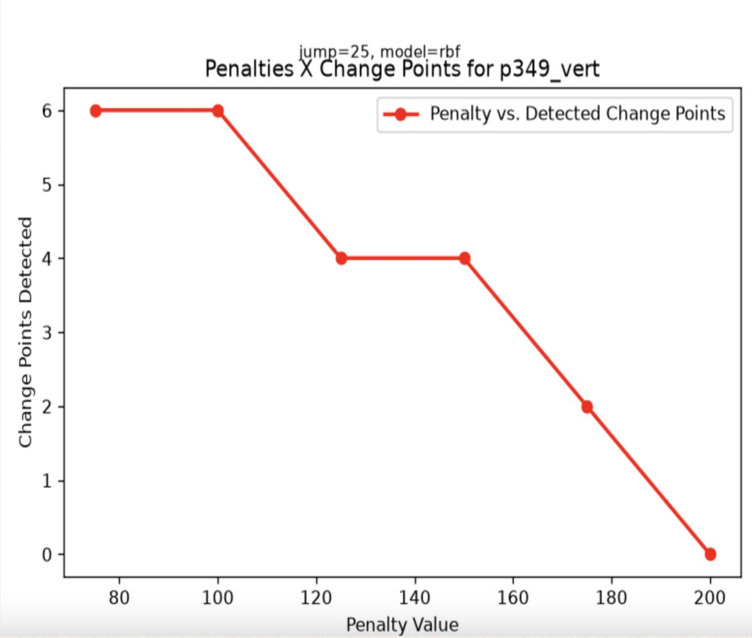
Changepoint Detection

Changepoint Detection (Brief)

Method

- Pruned exact linear time (PELT) algorithm
- INPUT: Time series, penalty

  OUTPUT: Optimal changepoint vector
- Iterative elbow-method (visual) penalty selection
- Tracks possible equipment changes, position offsets caused by earthquakes, etc



Example Penalty Score Testing Graph

04

Uncertainty Quantification

Does OLS's stated velocity uncertainty (σ^2_{OLS}) reflect reality, given the noise isn't white?

Key Motivating Papers

To quantify models' underestimation of formal uncertainty, I combine methods from two distinct fields:

Orbit Determination/Space Domain Awareness and **Geodesy**.

Bos Tero, Machiel Simon, et al. "Introduction to Geodetic Time Series Analysis." Geodetic Time Series Analysis in Earth, Springer Verlag, Aug. 2019.

Working Group on Covariance Realism. (n.d.). Covariance and Uncertainty Realism in Space Surveillance and Tracking (A. B. Poore, J. M. Aristoff, & J. T. Horwood, Eds.). Air Force Space Command Astrodynamics Innovation Committee.

OLS v.s. GLS

Overview

To verify whether OLS or GLS's stated uncertainty is more accurate, we need a **GROUND TRUTH** to verify against (in my case, ground truth is simulated).

We use Monte Carlo to generate N = 500 synthetic series with a predetermined VELOCITY, and some synthetic noise (all of which exhibiting the **same noise model** but with **different noise values in each epoch**).

OLS v.s. GLS cont.

Overview

Each model should be aware of its own spread (spread ~ uncertainty). So we may calculate formal σ^2 for each simulation.

Per Poore et al, we then calculate an **uncertainty metric** describing the consistency of uncertainty of the state (formal variance of velocity estimates) against a known truth state. It is the squared Mahalanobis Distance metric.

OLS v.s. GLS cont.

Overview

Notably, Mahalanobis distance is for multivariate cases. However, I am estimating a single parameter's uncertainty (df = 1). It follows that the squared Mahalanobis metric collapses to z^2

If the formal OLS/GLS σ^2 values were accurate, **the distribution of metrics should follow a χ2 distribution** with df = 1. A Kolmogorov χ2 Goodness of Fit is performed to test the null hypothesis: the empirical distribution conforms to the theoretical χ2 distribution.

The following slides will provide the mathematical reasoning behind the process.

Step-by-Step Walkthrough

1. Choose a station and direction. E.g. P349_east
2. Find the white and power law components of the noise model

The noise model (covariance matrix C) is written:

$$\mathbf{C} = \sigma_{pl}^2 \mathbf{J}(\kappa) + \sigma_w^2 \mathbf{I}$$

To find σ² for the power-law component, I find where sampling frequency = frequency. This is in accordance with the equation for power-law noise

$$P(f) = P_0 \left(f / f_s\right)^{\kappa}$$

To find σ² for the white component, I look at the power of each PSD graph at the flattening point (~5 cycles/yr). See "Spectral Analysis" slide for context.

Step-by-Step Walkthrough

3. Calculate J

H’ = transpose(H)

J = HH’

H is a lower triangular matrix computed from a recurrence relation:

$$h_i = (i - \frac{\kappa}{2} - 1)\frac{h_{i-1}}{i} \quad \text{for } i > 0$$

With H_0 = 1. K is retrieved from the P349_east’s fitted PSD line.

With this information I am now able to construct the parametric noise model, C.

Using C, I may generate N=500 synthetic Monte Carlo simulations with this noise model.

Step-by-Step Walkthrough

4. Construct N = 500 noise series, w

w = H**v**

H in this context: a noise filter applied to some randomly generated i.i.d series **v**.

w: "some error series with the same noise model as the original data."

Essentially:

$$\mathrm{cov}(w) = \mathrm{cov}(\boldsymbol{\sigma^2}H\mathbf{v}) + \mathrm{cov}(\boldsymbol{\sigma^2}\mathbf{I}) = \boldsymbol{\sigma^2}H\mathrm{cov}(\mathbf{v})H' + \boldsymbol{\sigma^2}\mathbf{I} = \boldsymbol{\sigma^2}HH' + \boldsymbol{\sigma^2}\mathbf{I} = C$$

By generating 500 unique i.i.d series **v,** I may construct 500 unique w error series. H is fixed per station.

Step-by-Step Walkthrough

5. Construct N = 500 synthetic series

Synthetic_series = XB + **w**

The deterministic part of the synthetic series, XB is constructed with pre-determined (ground-truth) parameter values. In other words, a priori, I set the true station velocity, semi-annual parameters, and intercept.

6. Fit OLS and GLS to all 500 series

OLS: B = (X'X)^(-1)X'y V.S. GLS: B = (X'C^(-1)X)^(-1)XC^(-1)y

Instead of computing C^-1 for GLS, I use the Cholesky decomposition which utilizes a unique, lower triangular matrix L, whitening all correlated variables and reducing imprecision.

Step-by-Step Walkthrough

7. Calculate formal σ^2 for OLS and GLS

OLS / GLS: $\sigma^2 = (1/n) * \Sigma(y\_synthetic - y\_predicted)^2$

OLS: $\sigma^2\_formal = \sigma^2 I$

GLS: $\sigma^2\_formal = \sigma^2 C$

8. Calculate uncertainty realism metrics

For a single point only (not a series/array):

GLS_metric = $(B\_predicted - B\_true)^2 / \sigma^2\_formal = z^2$ (same process for OLS)

Note: Mahalanobis² = $(x-y)S^{-1}(s-y)'$ → For multivariate problems. But in 1D, this is essentially “dividing some squared distance by variance,” which is what the uncertainty metric is.

Step-by-Step Walkthrough

9. Kolmogorov Goodness of Fit Test

Utilizes CDFs of the uncertainty metrics. Returns several statistics, including the statistic, which is the single greatest distance from theoretical CDF, and the P-Value

Here are the results of the test:

TEST RESULTS	Statistic	P - VALUE
GLS	0.00889	0.865219 (Passes)
OLS	0.57718	0.0 (Fails)

Recall H_null: The empirical distribution conforms to the theoretical χ2 distribution.

For OLS, we reject the null. OLS fails to conform to the theoretical distribution, and therefore fails to accurately report its formal uncertainty.

05

Additional Information

Project Script N^2 Chart

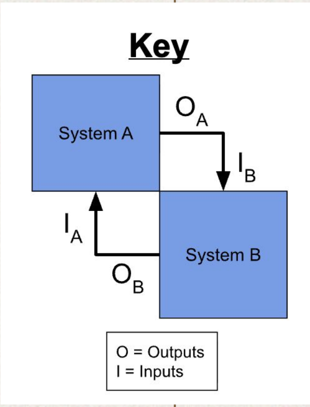
Raw Data

clean.py	Cleaned station data		
	signal_decomposition.py	Alphas (slopes) Betas X matrices Residuals Cleaned station data	
		changepoint_detection.py	All aforementioned + changepoints
			covariance_realism.py

PSD Graphs

Penalty Testing Graphs

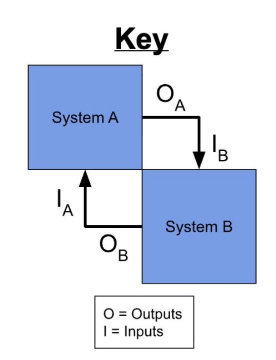
Metric Graphs



Covariance_realism.py N^2 Chart

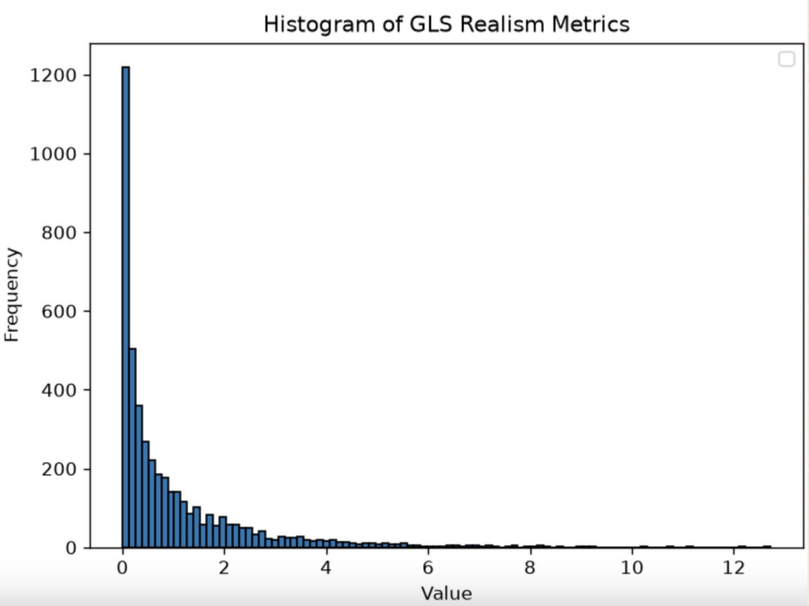
Station, direction

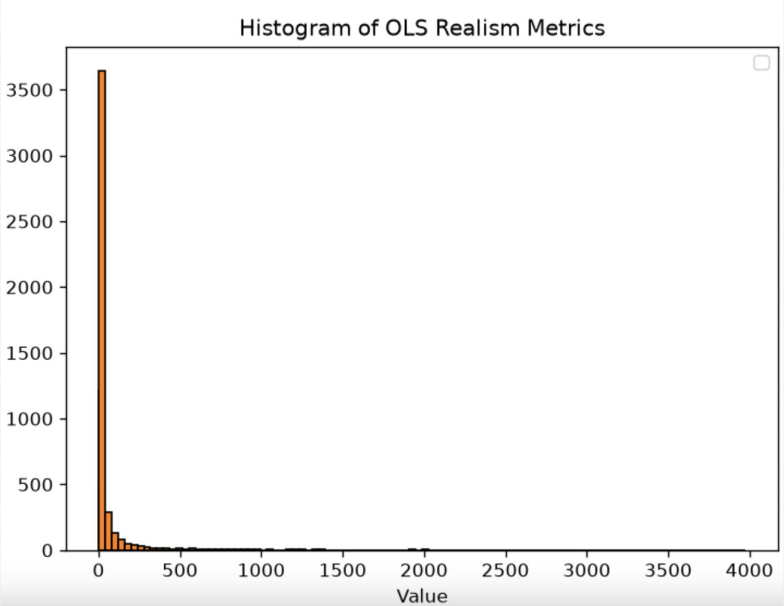
Create_gen_power_law_cov_matrix				• H • J		
	fit_psd_line		• slope • intercept			
		power_at_freq	• σ^2_white • σ^2_PL			
	• station • direction	• freq_w, freq_pl • Slope • intercept	**find_char_σ^2**	• σ^2_white • σ^2_PL		
				create_parametric_C	• C • slope • intercept	
• station • k					**generate_Monte_Carlo**	• Synthetic series • Velocity_true • C • Direction, station
						fit_LS_models



GLS/OLS Metrics

Realism Metric Distributions





“Value”: Covariance Realism Metric - Normalized distance from true variance

Project Limitations / Next Steps

- Trajectory model diagnostics (e.g., multicollinearity, predictor order) not performed → To preserve scope
  - However, model was carried from literature. Few drawbacks anticipated.
- Validated against simulated truth rather than verified truth → truth is inaccessible
  - The model, therefore, tests internal consistency.
- Changepoints not verified against true earthquake logs → To preserve scope
  - No direct effect (see next point)
- Outliers and Changepoints not removed / segmented → To preserve scope
  - As a result, OLS-based alpha is based upon the assumed noise model, not a corrected noise model
- Original data's noise was not tested for normality
  - The model, assumes normality, and therefore tests internal consistency only.

Thanks

Any more questions?
cvmomoh@ucdavis.edu
(408) 921-0045
Github
LinkedIn